

4th year project presentation

Dr Oliver Hadeler
February 2024

Details

Week 39 (23 & 24 May)

In-person parallel sessions

0.5 – 1h depending on group size

Content

1. Preparation
 - a) Topic
 - b) Slides and messages
 - c) Examples
 - d) Length
2. Presentation
3. Q&A

First thoughts

Find out

1. What is the purpose of the meeting?
2. Who is the audience?
3. How long should the presentation be?

The topic

Think about **one single idea** of your talk

What is your main idea?

Why would I want to listen to you?

The right level

Talk about something that interests you

Talk at a level adequate for the audience

One proposition per slide

Every slide has a proposition – tell the audience
what you will talk about

Introduce **one single idea** per slide!

Each slide conveys a message

Text, figures and tables should convey a message

not

to show “that you have done something”.

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Ask yourself: “What will I say about this graph?”
Do I need it?

Each slide conveys a message

Text, figures and tables should convey a message

not

to show “that you have done something”.



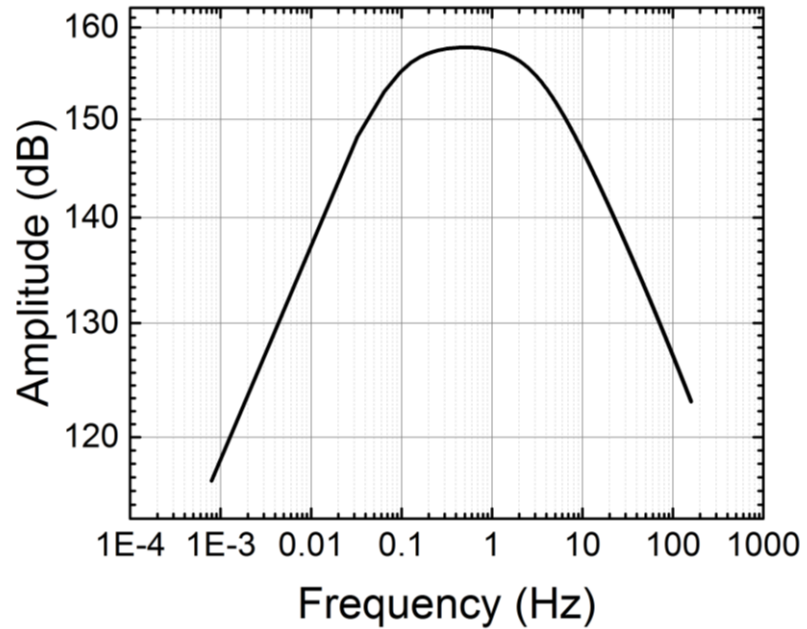
Ask yourself: “What will I say about this graph?
Do I need it?”

“Here is another set of results...”

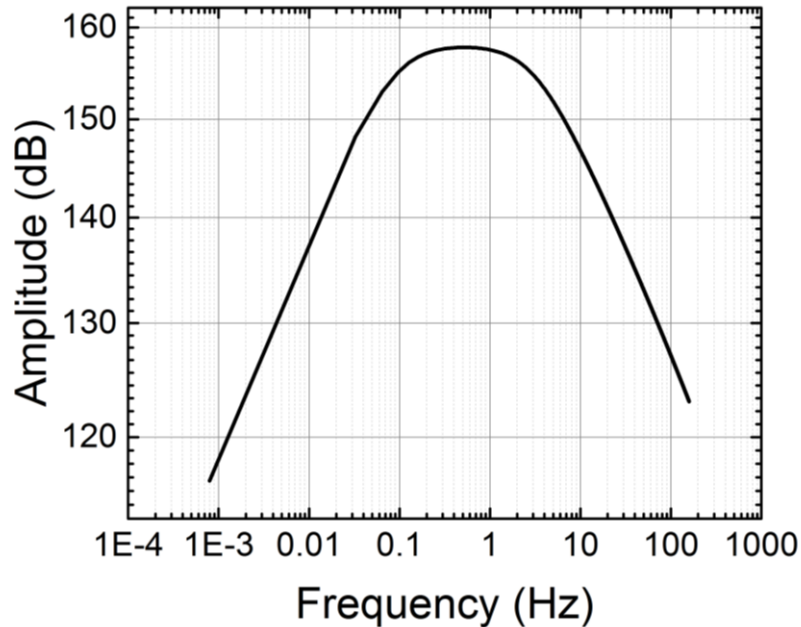
Some examples

Let us look at some real examples

2nd design

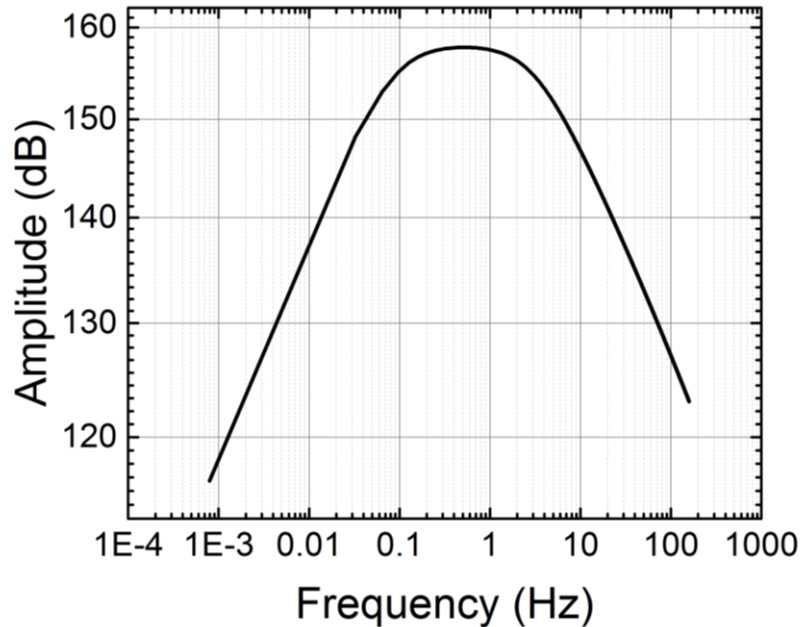


2nd design



- What is a “2nd design”?
- What is the graph showing us? Title?
- What are the important points on the graph?

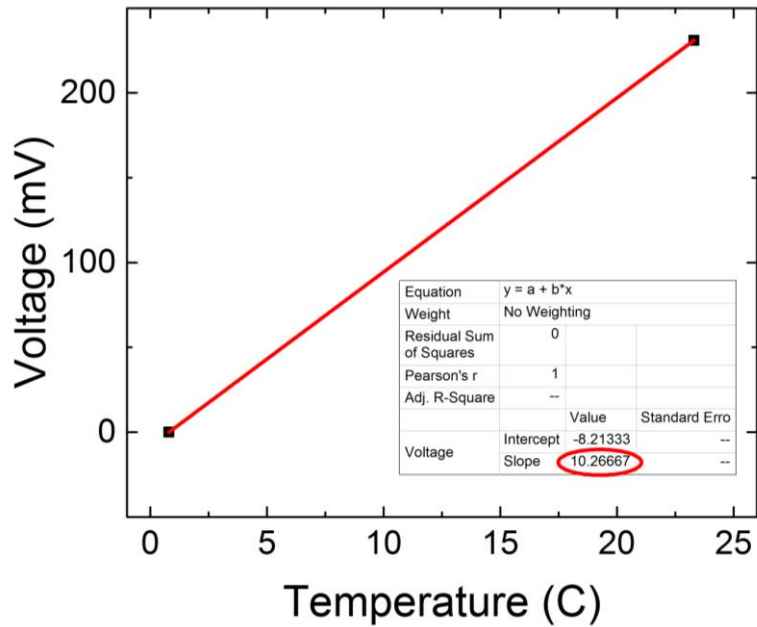
2nd design



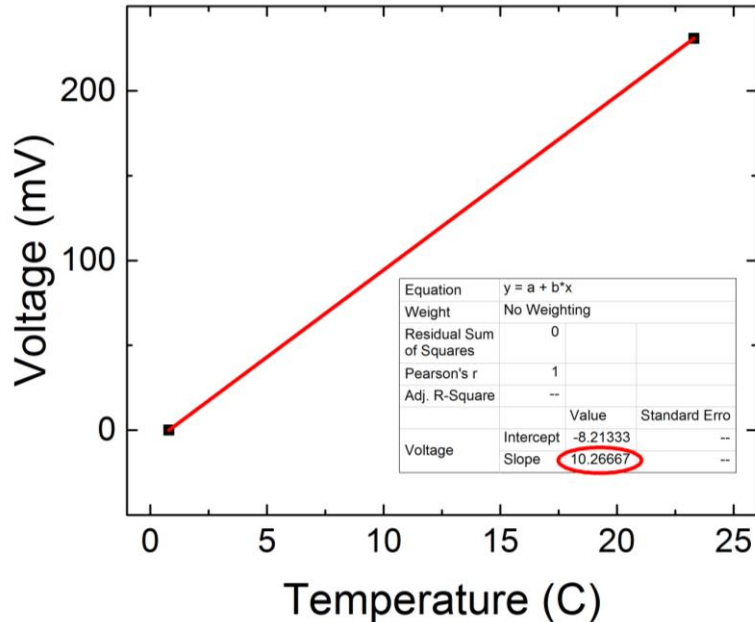
- What is a “2nd design”?
- What is the graph showing us? Title?
- What are the important points on the graph?

Good:
It is not a screenshot

No calibration needed

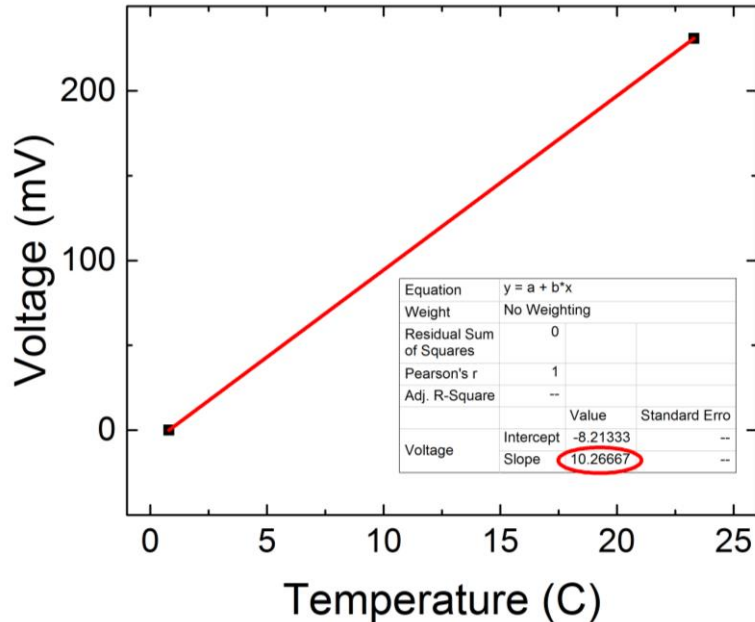


No calibration needed



- A straight line through two points!
- Do you need all the statistical information?

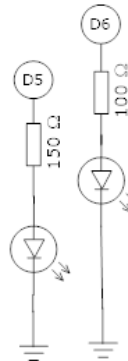
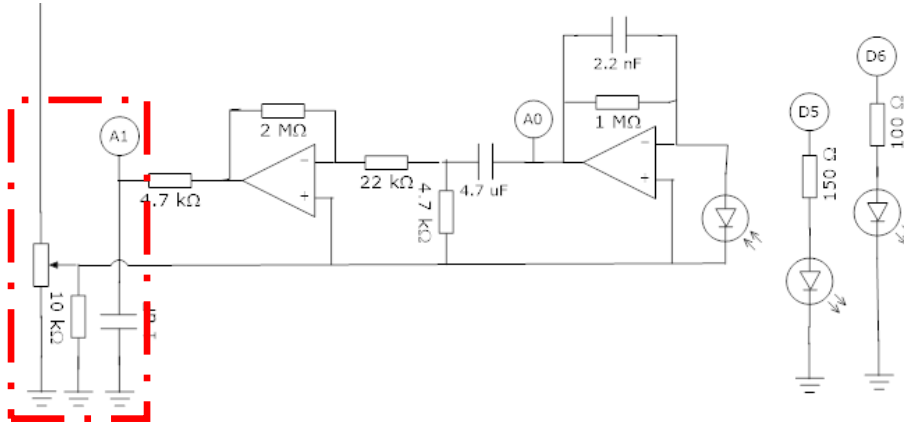
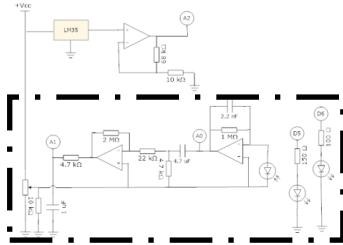
No calibration needed



- A straight line through two points!
- Do you need all the statistical information?

Good:
Highlighted the important information

Circuit design

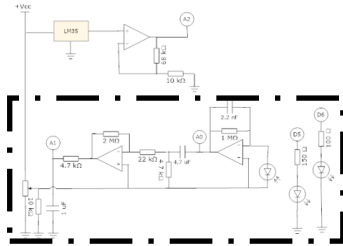


72.34 Hz

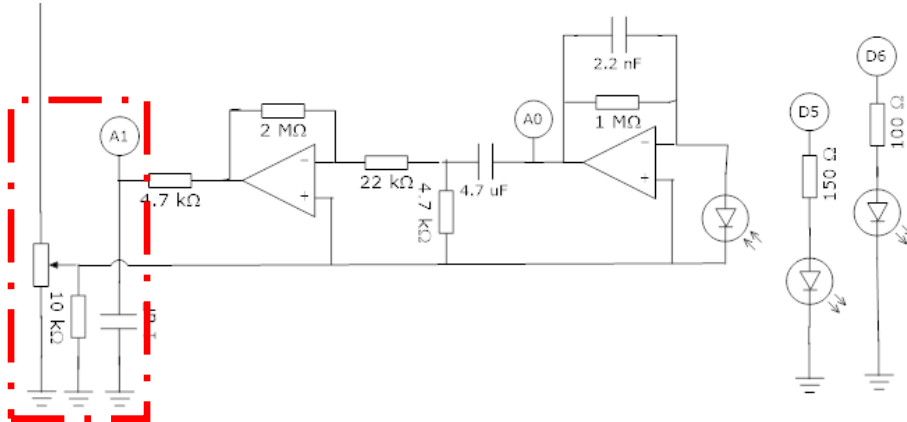
0.72 Hz

33.86 Hz

Circuit design



- Lack of focus
- What are the frequencies?
- Two decimal places?

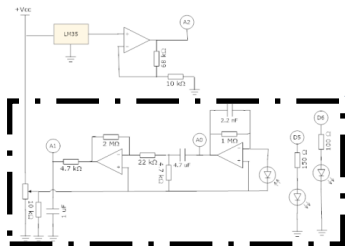


72.34 Hz

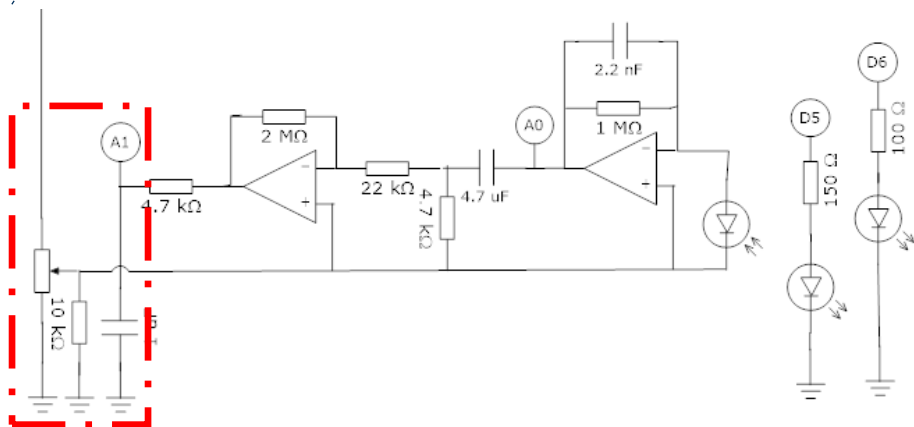
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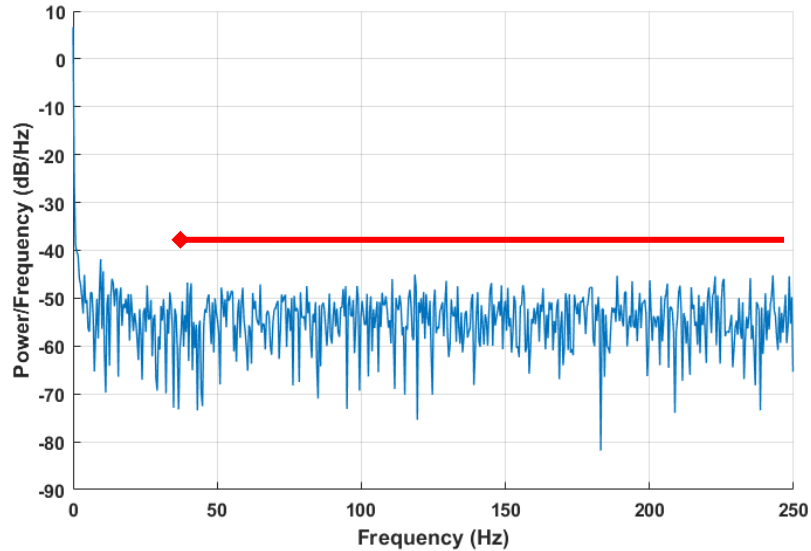
72.34 Hz

0.72 Hz

33.86 Hz

Good:
Slide title

Circuit design

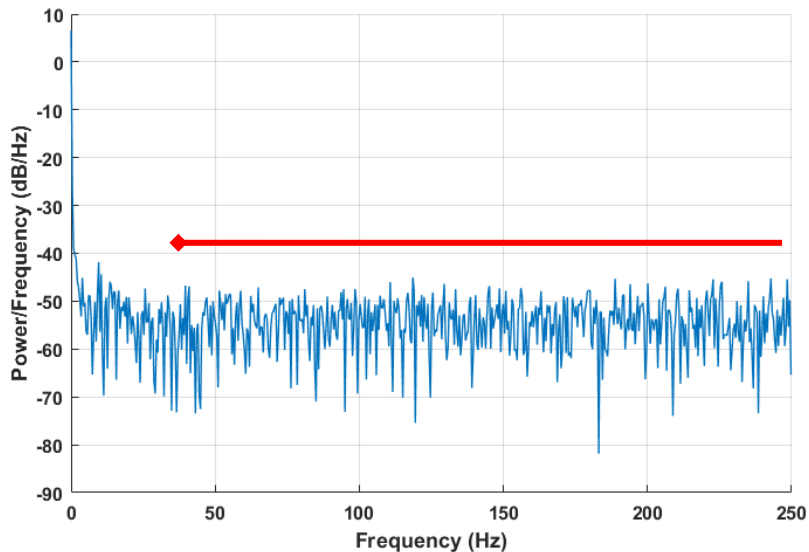


72.34 Hz

0.72 Hz

33.86 Hz

Circuit design



- Font size too small
- What is the red line?
- Frequency scale

72.34 Hz

0.72 Hz

33.86 Hz

Adjusting a 9600 Baud Rate

- Check the datasheet for PIC16F1716

TABLE 31-3: BAUD RATE FORMULAS

Configuration Bits			BRG/EUSART Mode	Baud Rate Formula
SYNC	BRG16	BRGH		
0	0	0	8-bit/Asynchronous	$F_{osc}/[64 (n+1)]$
0	0	1	8-bit/Asynchronous	$F_{osc}/[16 (n+1)]$
0	1	0	16-bit/Asynchronous	
0	1	1	16-bit/Asynchronous	$F_{osc}/[4 (n+1)]$
1	0	x	8-bit/Synchronous	
1	1	x	16-bit/Synchronous	

- n is 16 bits: SPBRGH : SPBRGL (we use 0 : 25)
- Using the formula : baud rate = 9615 (0.16% error)

Adjusting a 9600 Baud Rate

- Check the datasheet for PIC16F1716

TABLE 31-3: BAUD RATE FORMULAS

– Which information do we need? – Target audience?				Baud Rate Formula
0	0	1	8-bit/Asynchronous	$F_{osc}/[64 (n+1)]$
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0	1	1	16-bit/Asynchronous	$F_{osc}/[16 (n+1)]$
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Adjusting a 9600 Baud Rate

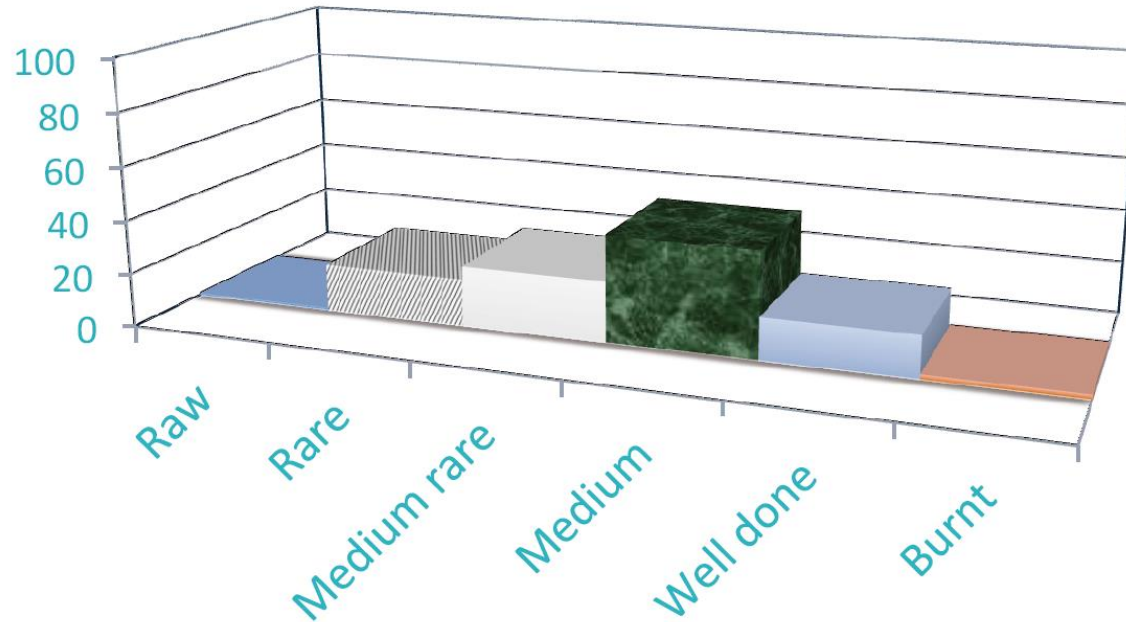
- Check the datasheet for PIC16F1716

TABLE 31-3: BAUD RATE FORMULAS

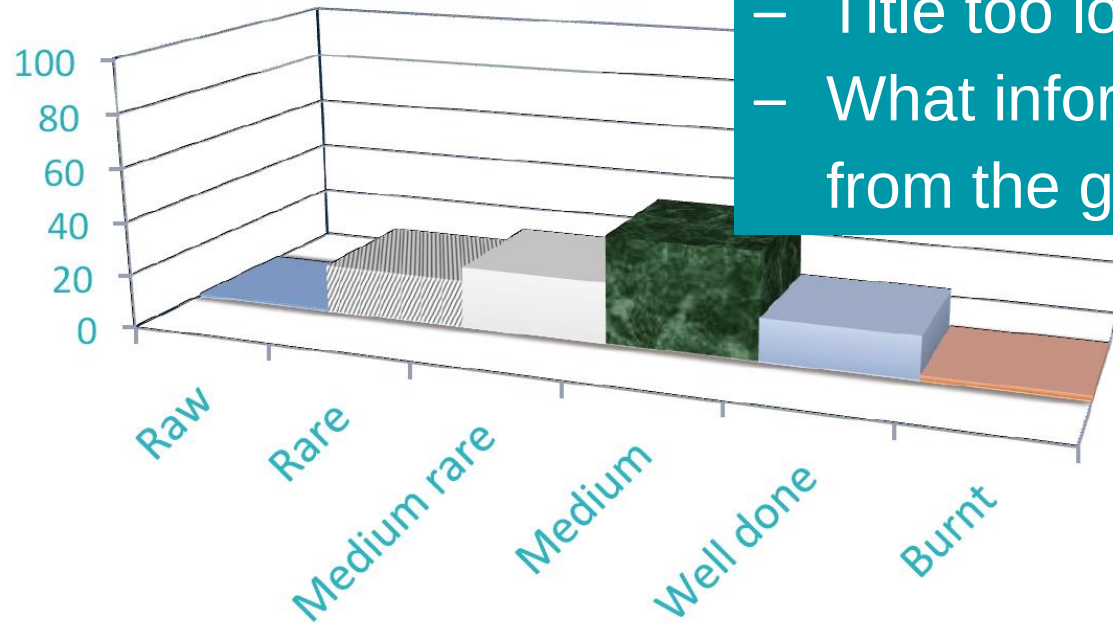
– Which information do we need? – Target audience?	Baud Rate Formula		
	$F_{osc}/[64 (n+1)]$		
Good: Source is acknowledged	$F_{osc}/[16 (n+1)]$		
	$F_{osc}/[4 (n+1)]$		

- n is 16 bits: SPBRGH : SPBRGL (we use 0 : 25)
- Using the formula : baud rate = 9615 (0.16% error)

Percentage response in a survey out of 278 individuals to the question: "How do you like your steak done?"

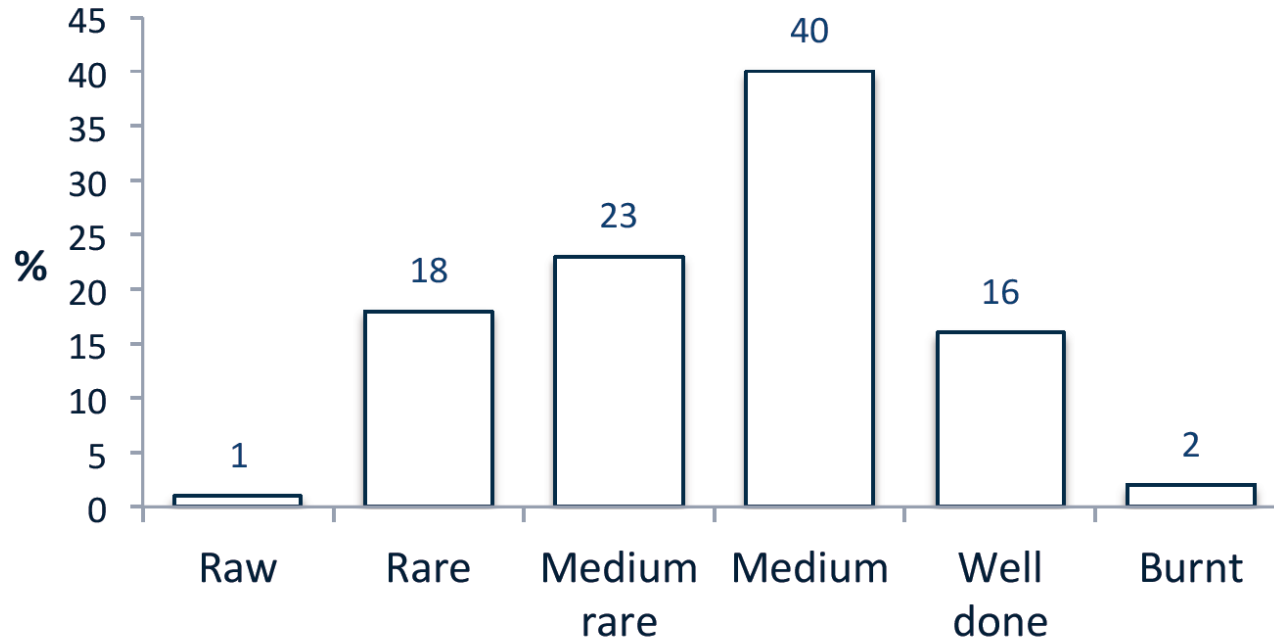


Percentage response in a survey out of 278 individuals to the question: "How do you like your steak done?"

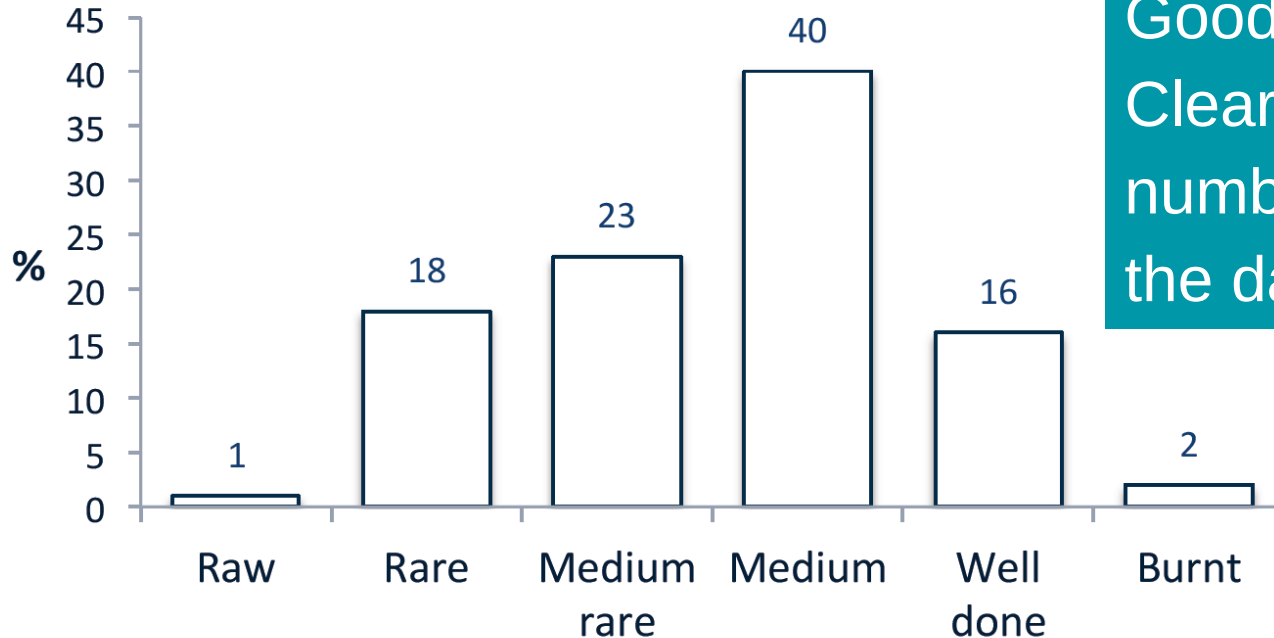


- Title too long
- What information do you gain from the graph design?

Percentage response in a survey out of 278 individuals to the question: "How do you like your steak done?"



Percentage response in a survey out of 278 individuals to the question: "How do you like your steak done?"



Good:
Clear graph,
numbers reinforce
the data

Presentation structure – similar to a report

1. Introduction
2. Theory
3. Methods
4. Results
5. Discussion
6. Conclusions
7. Summary of contributions

Presentation structure – similar to a report

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Think about the “flow”:

Sub-topic 1, Sub-topic 2, Sub-topic 3

Presenter 1: Introduction & theory

Presenter 2: Methods & results

...

Avoid too many switches between presenters

Presentation length

Calculation:

10min presentation per team member

5min Q&A per team member

You do not have to stick exactly to this individual timing, but overall duration is strict.

Q&A will be at the end of the presentation

Presentation length

Example:

Group of three students

30min presentation + 15min Q&A

Each student will contribute about
10min to the presentation

Number of slides

As a guide:

1min per slide

10min per student = maximum 10 slides per student

E.g. group of 3 students = no more than 30 slides

Better aim for ~20 slides

Presenting

- Speak clearly
- Address all points on the slide
- Look at the audience, do not read from the slide or your notes all the time

Presenting

Tell the audience what it is you will tell them

Tell it

Tell them what you have told them

Taking questions

- Look at the person asking the question
- Listen to the question
- Maybe repeat the question in your own words.
E.g. “you were asking if ...”
- Answer the question – examiners are interested in your work